

**VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



PROGRESS REPORT 1

Course: Machine Learning (CO3017)

Semester: 252 – **Academic year:** 2025 – 2026

Instructor: Dr. Truong Vinh Lan

Ho Chi Minh City, March 8

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1. GROUP INFORMATION

No.	Full name	Student ID
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2. TASK ALLOCATION

Phase 1: Problem Definition & EDA

While specific tasks were allocated to individual members to optimize progress, all team members actively contributed to every stage of the workflow through continuous discussion and mutual support, ensuring a collective understanding of the project's foundation.

Member	Responsibilities	Status
Phan Van Minh Tien	Allocate tasks, Set up the development environment; Implement core EDA scripts; Perform data loading and technical inspection of the feature space.	100%
Le Quang Thanh	Conducted research on alternative datasets; Evaluated the practical significance and clinical logic of the target variables.	100%
To Dang Duc Tai	Conducted research on alternative datasets. Evaluated the practical significance and clinical logic of the target variables.	100%
Nguyen Trong Anh Minh	Analyzed data distributions and identified outliers in key features like Glucose levels and BMI; Interpreted correlation matrices.	100%
Nguyen Thi Anh Thuy	Drafted the progress report; Translated technical findings into academic English; Formatted visualizations for the final document.	100%

3. PROJECT OVERVIEW

Selected Task: Machine Learning with Tabular Data.

Dataset: Stroke Prediction Dataset ([Kaggle](#)).

Objective: To build a traditional machine learning pipeline to predict the likelihood of a stroke based on clinical and demographic features.

4. DATASET JUSTIFICATION

The selected dataset satisfies the mandatory constraints defined in the project description:

- **Missing Values:** The BMI column contains 201 missing values, allowing us to implement and compare imputation techniques.
- **Categorical Values:** The dataset includes multiple categorical features (e.g., gender, work_type, smoking_status), requiring encoding strategies.
- **Sample Size:** With over 5,000 samples, the size is sufficient for a meaningful machine learning pipeline.

5. EXPLORATORY DATA ANALYSIS (EDA) RESULTS

- **Missing Data Analysis:** Using the missingno library, we identified that approximately 3.9% of the records lack BMI data. We plan to evaluate Median Imputation and KNN Imputation.

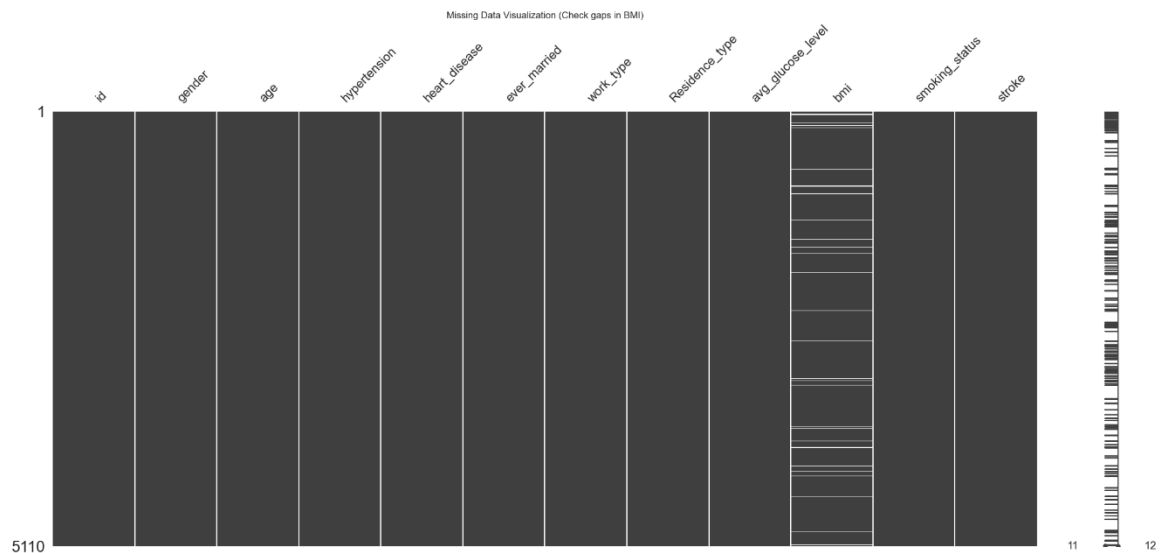


Figure 5.1: Missing data matrix highlighting the gaps in the BMI column.

- **Class Imbalance:** Initial visualization shows a significant imbalance, with only 4.87% of participants having suffered a stroke. This will guide our choice of evaluation metrics (F1-Score over Accuracy).

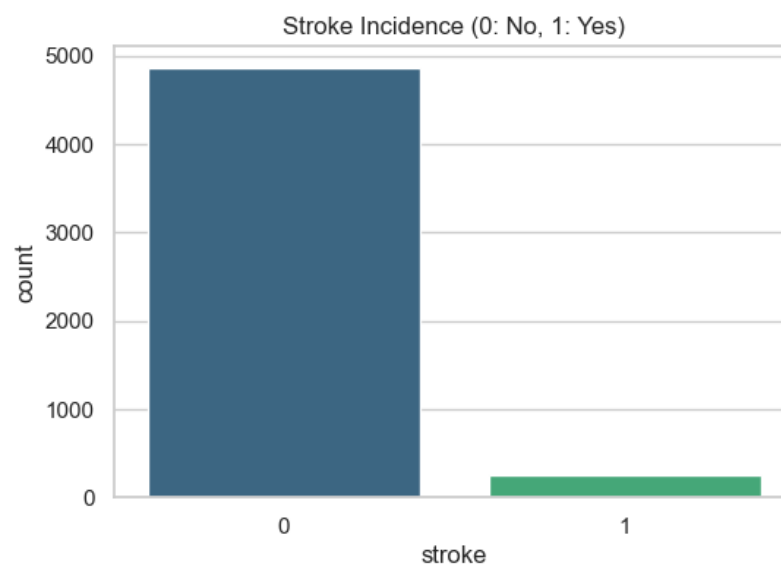


Figure 5.2: Distribution of the target variable showing significant class imbalance.

- **Feature Correlation:** Preliminary heatmaps indicate that age has the strongest positive correlation with stroke incidence.

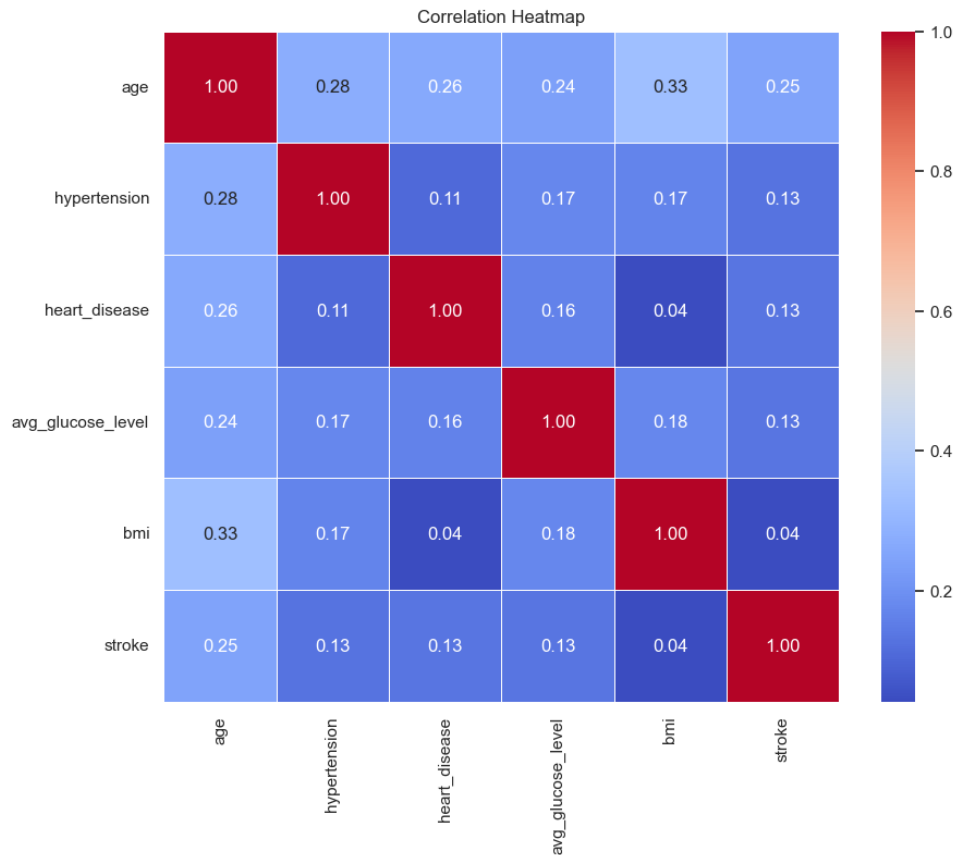


Figure 5.3: Correlation heatmap showing the relationship between physiological factors and stroke risk.

Numerical Feature Analysis (Outliers & Distributions): Our analysis using Boxplots and Histograms identified critical characteristics of the numerical features:

- **Average Glucose Level:** The data is significantly right-skewed with a large number of outliers exceeding the 200 mg/dL mark. These high values are clinically relevant as they represent diabetic or pre-diabetic states, which are strong risk factors for stroke.
- **BMI:** This feature also exhibits numerous outliers, with several values exceeding 45. Additionally, the presence of missing values in this column requires a robust imputation strategy (like KNN Imputer) to avoid introducing bias.

- **Age:** Unlike the other two, age shows a relatively uniform distribution with no significant outliers, making it a stable and powerful predictor in our feature set.

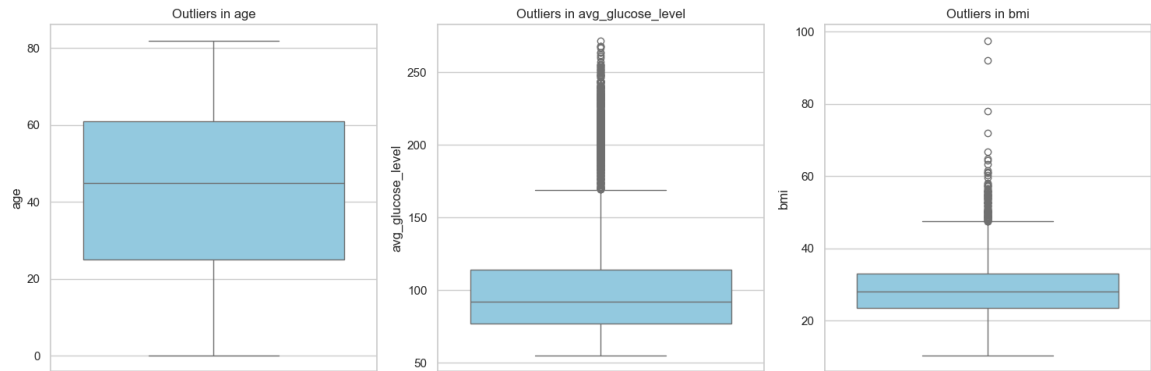


Figure 5.4: Boxplot analysis for outlier detection in numerical features (Age, Glucose, BMI)

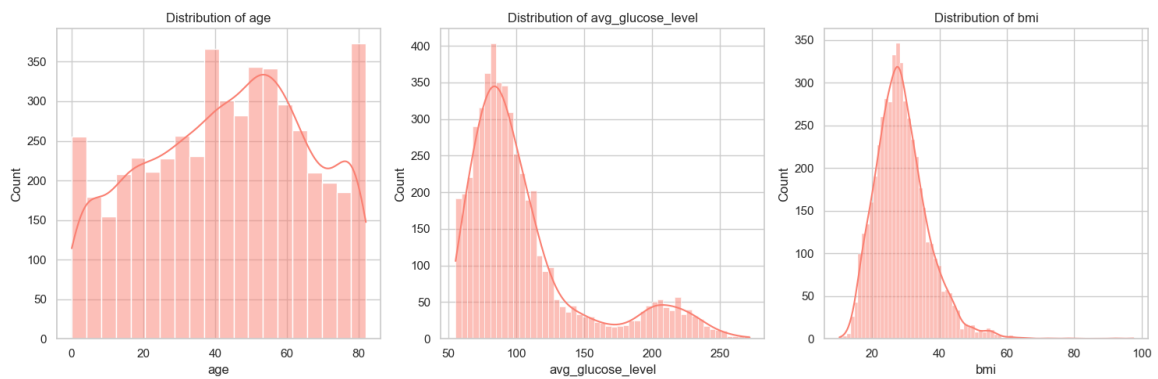


Figure 5.5: Distribution trends of numerical features highlighting the right-skewness of Glucose and BMI.

6. PROJECT STRUCTURE & REPOSITORY

We have established the required directory structure for the final submission:

- notebooks/: Contains the Exploratory Data Analysis (EDA) Jupyter Notebooks.
- modules/: Placeholder for Python modules for preprocessing and training.
- reports/: Stores progress reports and documentation.
- features/: Directory for saved feature vectors (e.g., .npy or .h5 files).

The project's source code, datasets, and documentation are managed via GitHub to ensure version control and collaboration.

Repository Link: <https://github.com/PhanVanMinhTien/ml-project>

7. NEXT STEPS

1. Implementation of the preprocessing module (Scaling, Encoding, Imputation).
2. Initial baseline testing with Random Forest.
3. Evaluation of PCA (Principal Component Analysis) for dimensionality reduction.